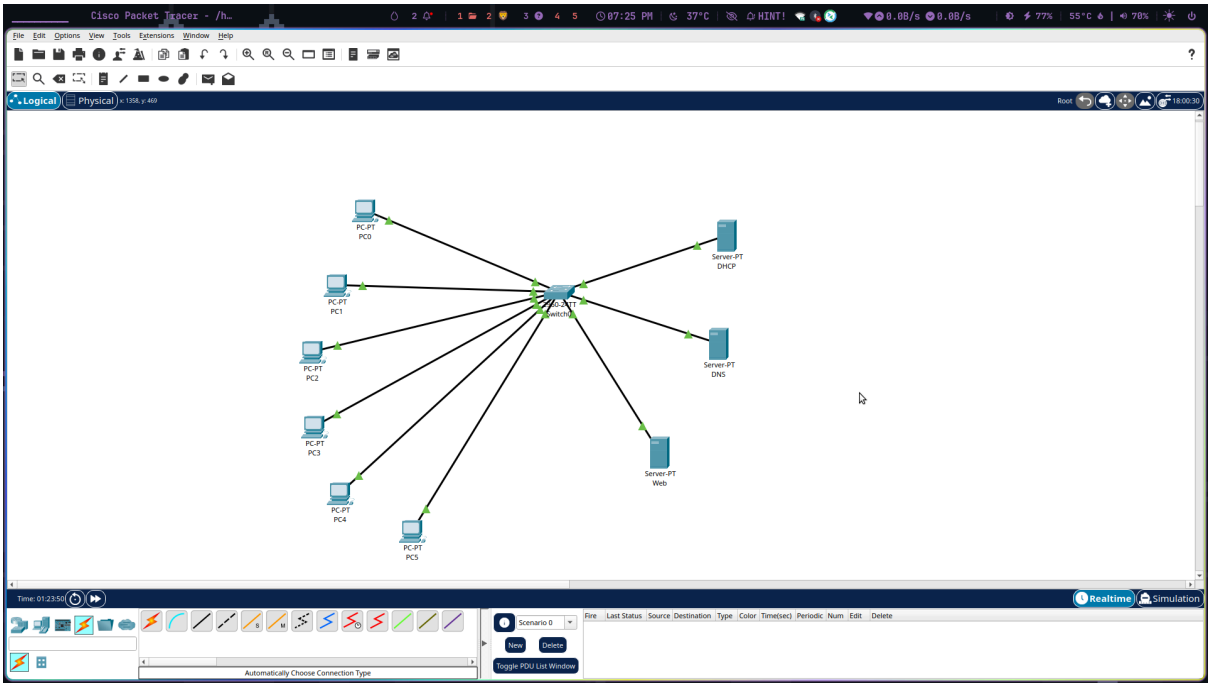
ASSIGNMENT 11
COMPUTER NETWORK
ID: 12341550

PART A : Topology Design

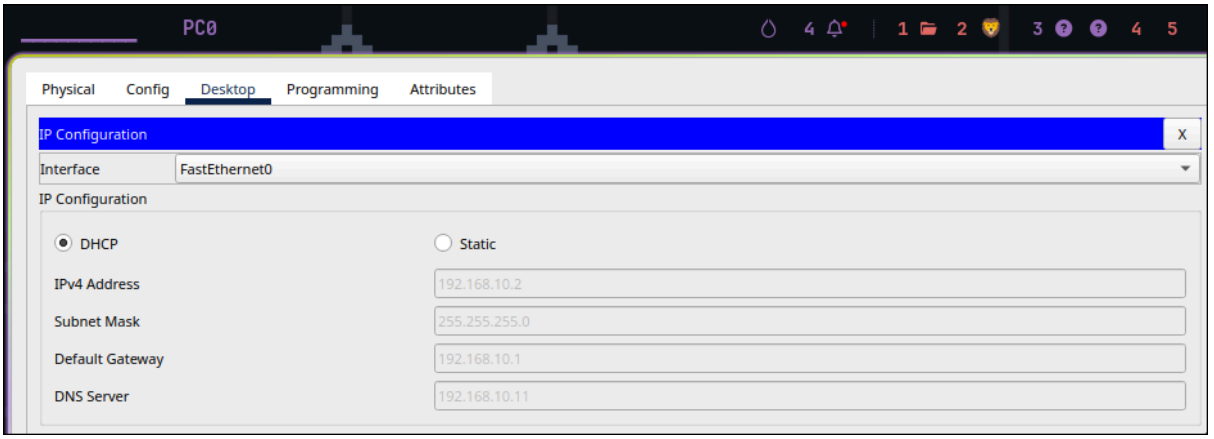
- Placed 1 Cisco 2960 Switch, 3 Servers (designated as DHCP, DNS, and Web), and 6 Student PCs into the logical workspace.
- Connected all end devices and servers to the central switch using Copper Straight-Through cables to establish a single Local Area Network (LAN).
- Verified physical connectivity by ensuring all port link lights turned green.



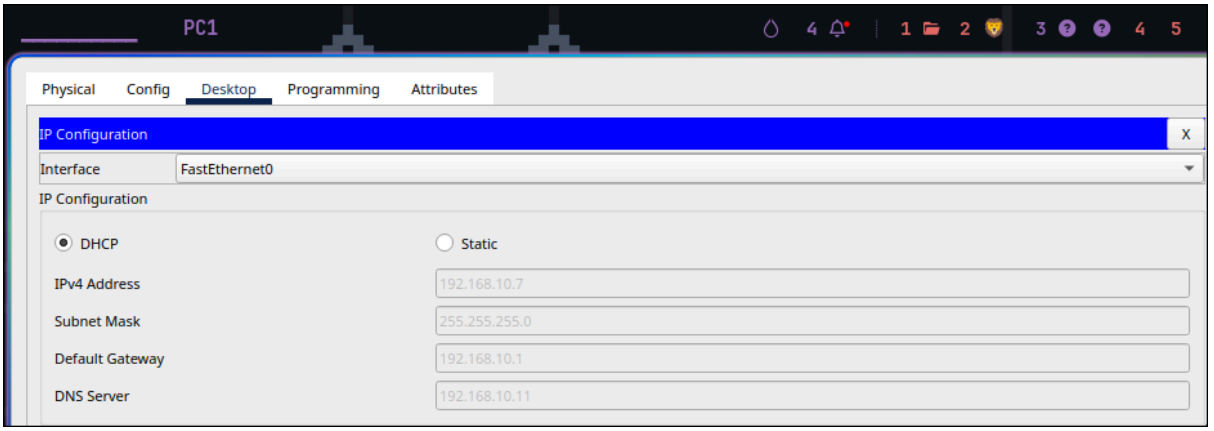
PART B : Configuration

1. **Configure all PCs via DHCP.**

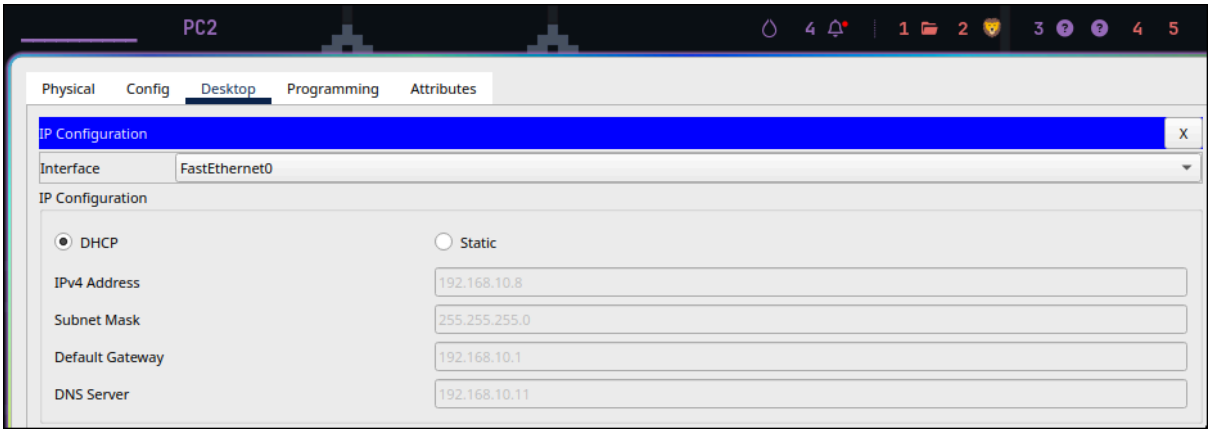
    - Accessed the IP Configuration utility on the Desktop tab for each of the 6 PCs (PC0 to PC5).
    - Switched the IPv4 configuration mode from Static to DHCP.
    - Verified that each PC successfully broadcasted a DHCP request and obtained an IP address from the pool, along with the correct Subnet Mask (255.255.255.0), Default Gateway (192.168.10.1), and DNS Server address (192.168.10.11).


PC0
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.10.2
Subnet Mask
255.255.255.0
Default Gateway
192.168.10.1
DNS Server
192.168.10.11

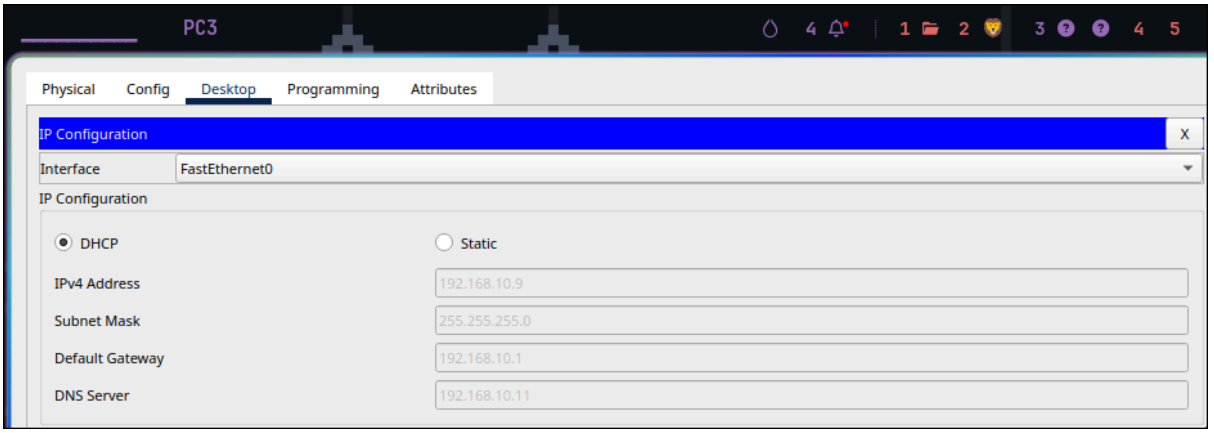
PC0


PC1
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.10.7
Subnet Mask
255.255.255.0
Default Gateway
192.168.10.1
DNS Server
192.168.10.11

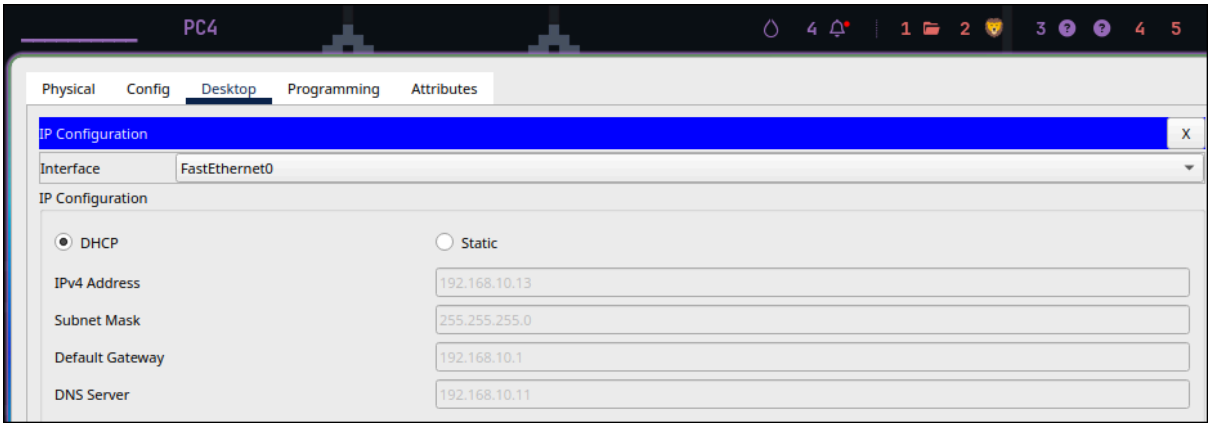
PC1


PC2
Physical
Config
Desktop
Programming
Attributes
IP Configuration
X
Interface
FastEthernet0
IP Configuration
DHCP
Static
IPv4 Address
192.168.10.8
Subnet Mask
255.255.255.0
Default Gateway
192.168.10.1
DNS Server
192.168.10.11

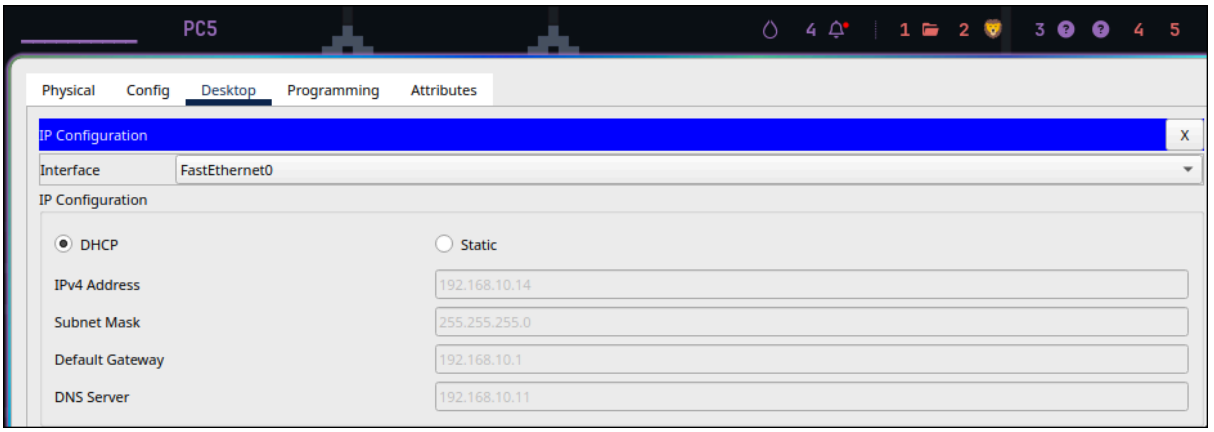
PC2



PC3



PC4

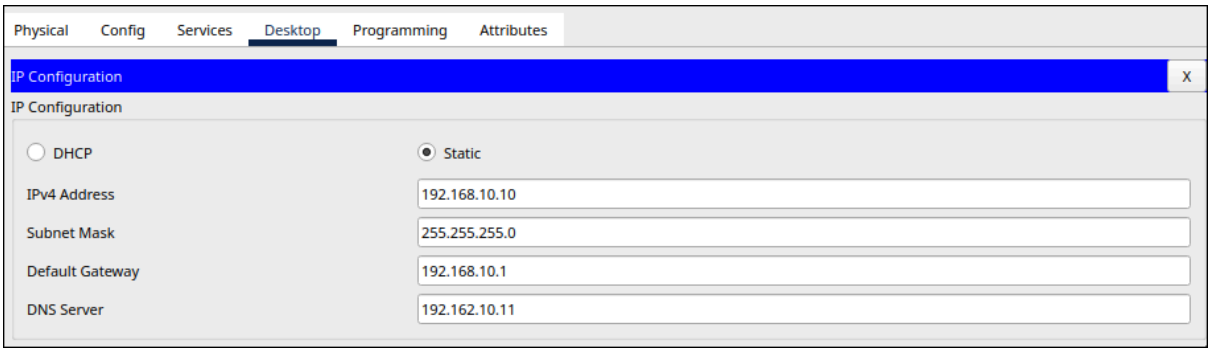


PC5

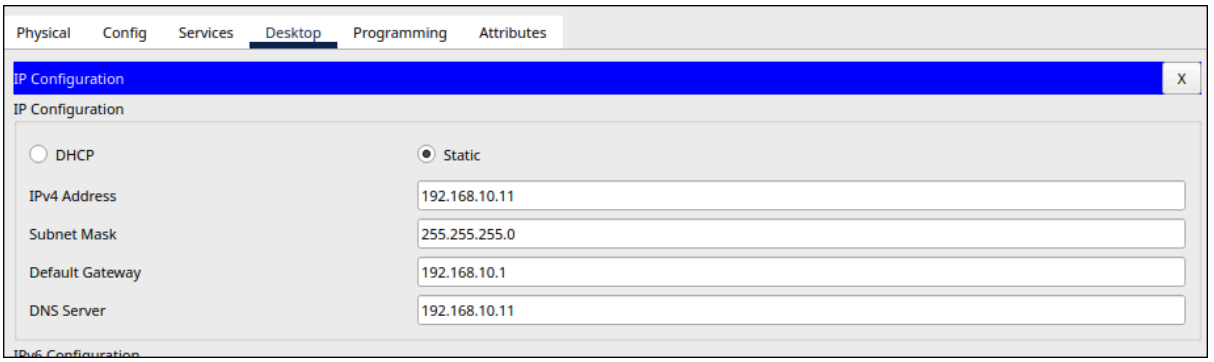
2. **Configure all Servers with Static IP.**

   - Manually assigned static IPv4 parameters to the three infrastructure servers to ensure permanent addresses for reliable network services.
   - **DHCP Server:** Assigned IP `192.168.10.10`
   - **DNS Server:** Assigned IP `192.168.10.11`

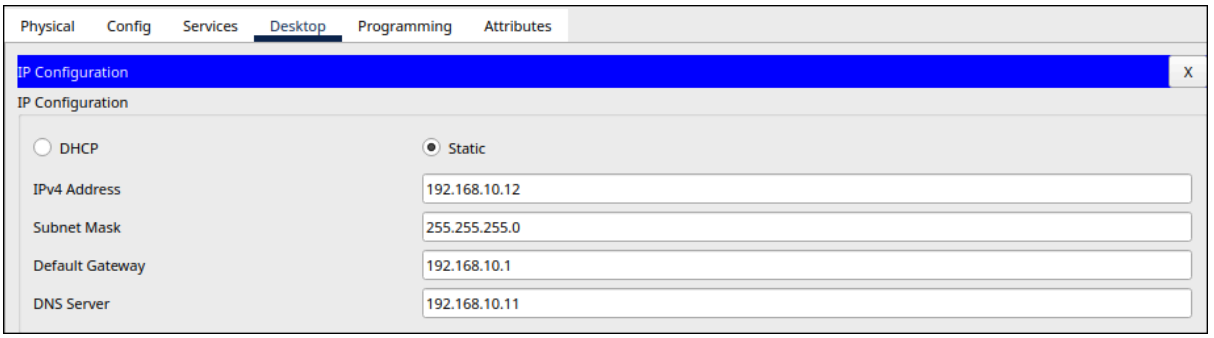
- **Web Server:** Assigned IP `192.168.10.12`



DHCP



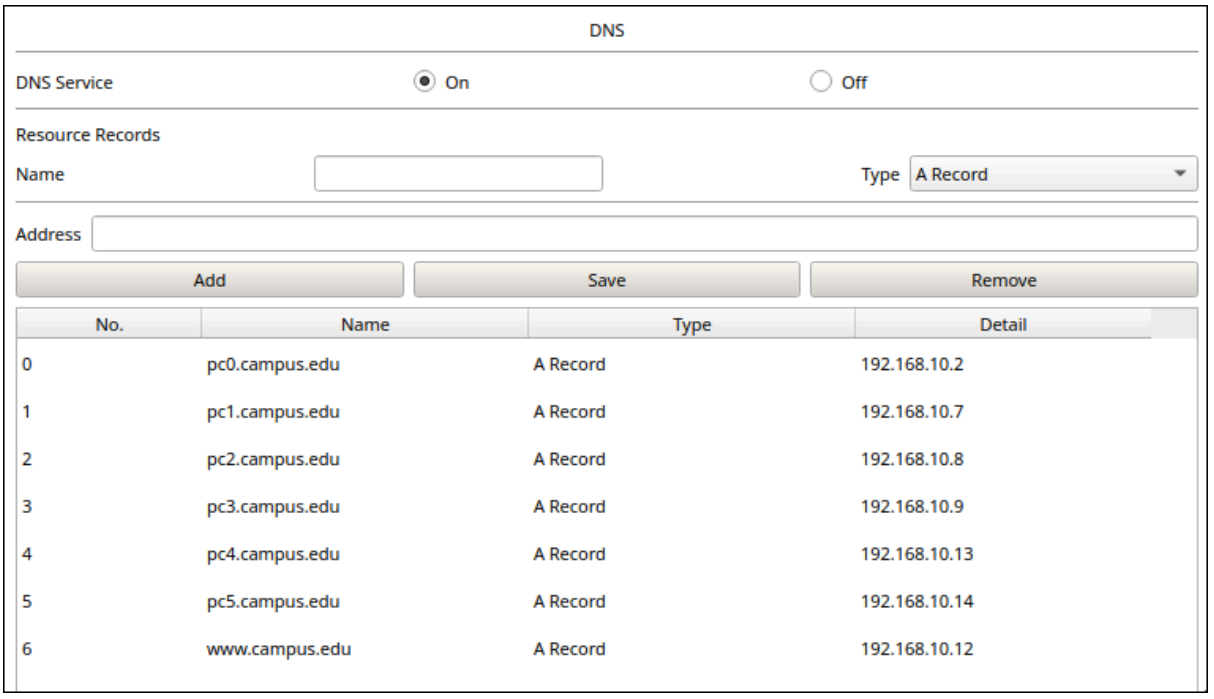
DNS



Web

3. **Assign IP and name mapping in the DNS Server correctly.**

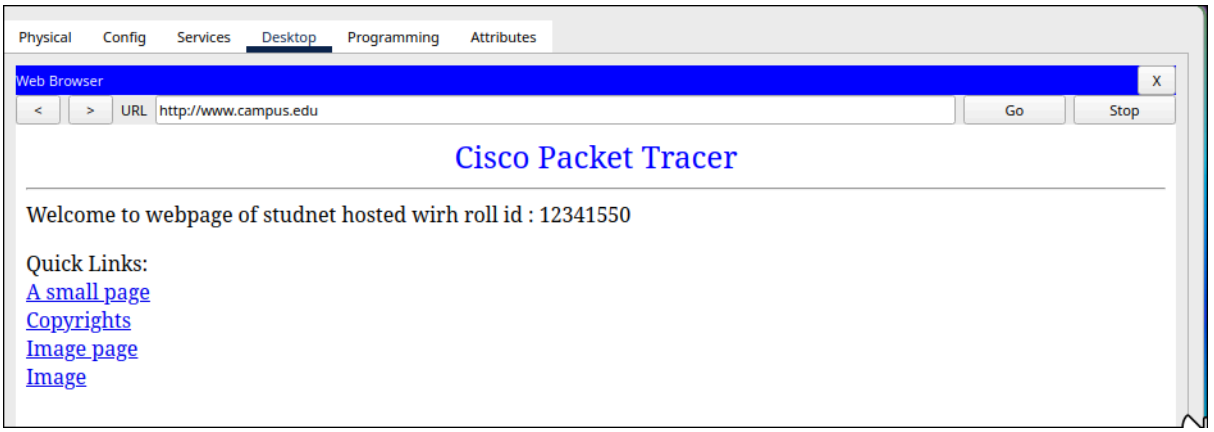
- Enabled the DNS Service on the DNS Server.
- Created 'A Records' linking human-readable domain names to their respective static or dynamically assigned IP addresses.
- Mapped the main site `www.campus.edu` to the Web Server (`192.168.10.12`).
- Created individual records for the student computers (e.g., mapping `pc0.campus.edu` to its DHCP-leased IP of `192.168.10.2`).



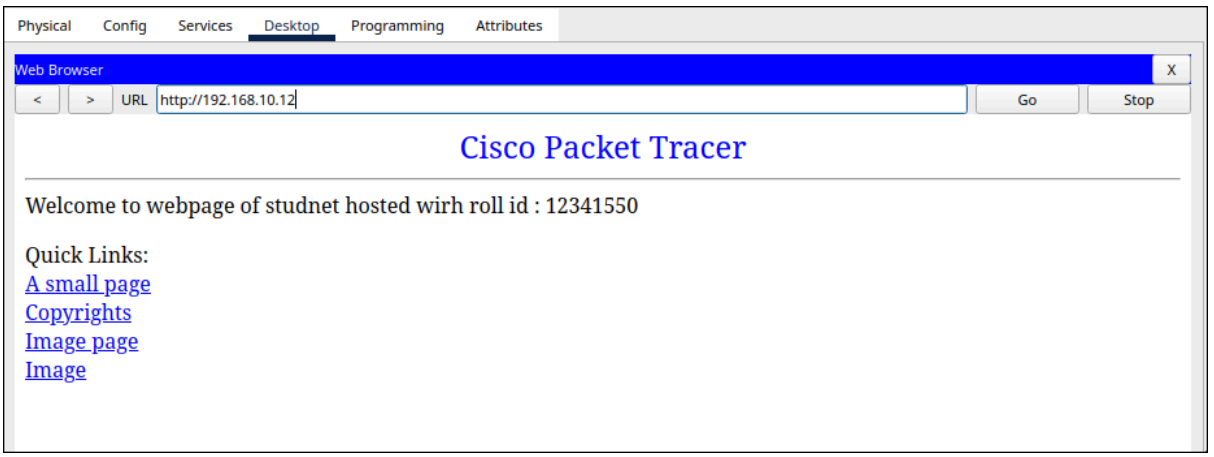
IP and name mapping

4. **Configure the Webserver Demo Website to work with both the IP address and the domain name.**

    - Enabled the HTTP/HTTPS services on the Web Server.
    - Edited the `index.html` file using the File Manager to display a custom digital campus welcome message, uniquely identifying the page with my student roll number (12341550).



with domain



with ip

PART C : Verification

1. **Show that all PCs receive IP configuration dynamically using DHCP (IP, subnet mask, default gateway).**

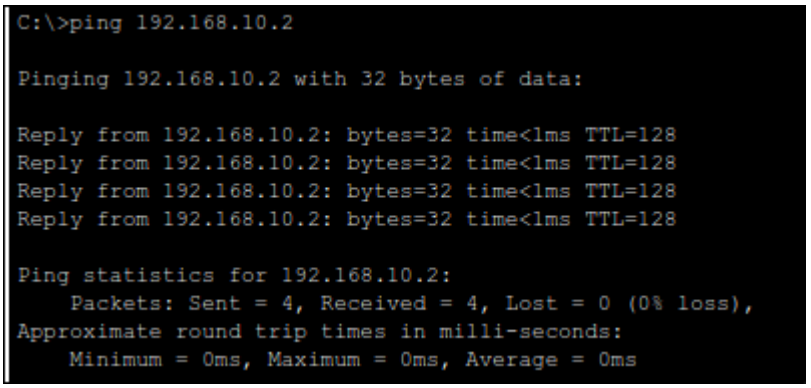
   As demonstrated in the Part B screenshots, the network adapters of all 6 PCs successfully leased addresses from the `192.168.10.0/24` network. This confirms the centralized DHCP server is actively managing IP allocation, satisfying the IT department's requirement to eliminate manual IP conflicts.

2. **Ping between at least two PCs using:**

   Conducted ICMP echo requests (ping) between end devices to verify end-to-end connectivity across the switch.

   a. **IP address**

      Successfully pinged PC0 (`192.168.10.2`) from PC2, and PC3 from PC5, resulting in 4 successful replies and 0% packet loss.



pc2 -> pc0

```
C:\>ping 192.168.10.9

Pinging 192.168.10.9 with 32 bytes of data:

Reply from 192.168.10.9: bytes=32 time=1ms TTL=128
Reply from 192.168.10.9: bytes=32 time<1ms TTL=128
Reply from 192.168.10.9: bytes=32 time<1ms TTL=128
Reply from 192.168.10.9: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.9:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

pc5 -> pc3

b. Domain name

Successfully pinged `pc0.campus.edu` from PC2, and `pc3.campus.edu` from PC5. This proves that the DNS server correctly resolved the hostnames to their respective IP addresses before initiating the ping.

```
C:\>ping pc0.campus.edu

Pinging 192.168.10.2 with 32 bytes of data:

Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128
Reply from 192.168.10.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

pc2 -> pc0

```
C:\>ping pc3.campus.edu

Pinging 192.168.10.9 with 32 bytes of data:

Reply from 192.168.10.9: bytes=32 time<1ms TTL=128
Reply from 192.168.10.9: bytes=32 time<1ms TTL=128
Reply from 192.168.10.9: bytes=32 time<1ms TTL=128
Reply from 192.168.10.9: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.10.9:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

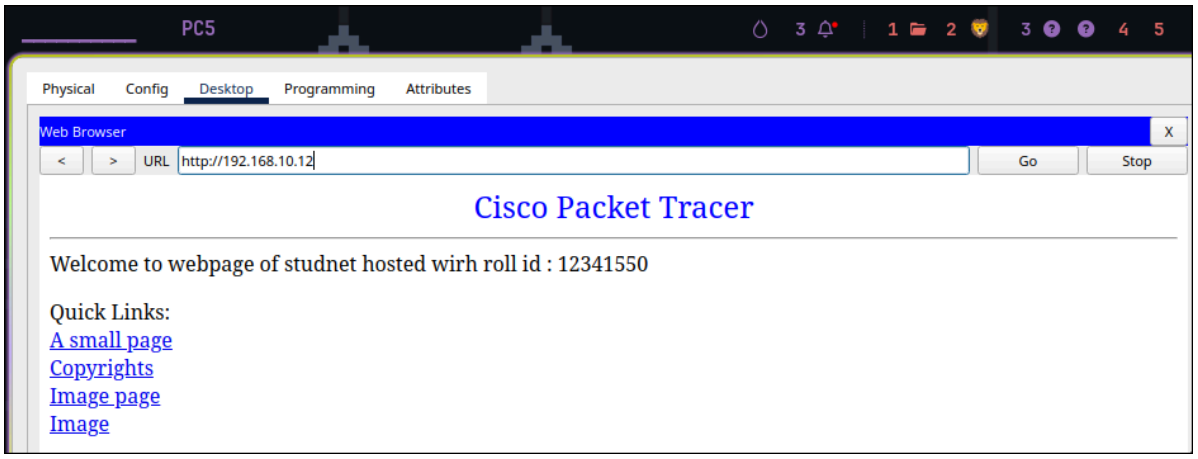
pc5 -> pc3

3. Access the web server:

Utilized the web browser application on a student PC (PC5) to retrieve the custom HTML page hosted on the internal Web Server.

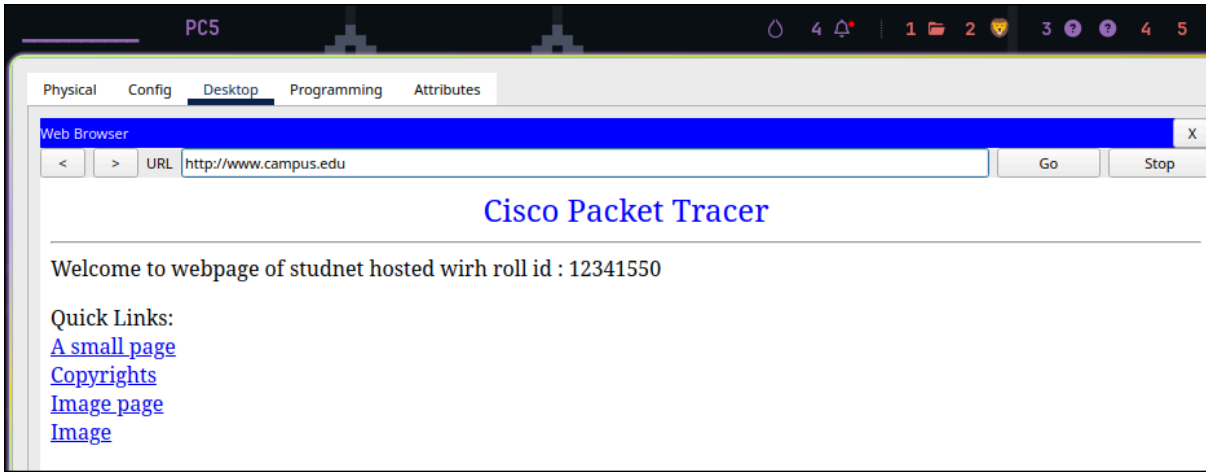
a. **Using IP address**

Navigated directly to `http://192.168.10.12`, which successfully loaded the customized digital campus demo page.



b. **Using domain name**

Navigated to `http://www.campus.edu`. The DNS server resolved the domain, successfully loading the identical demo page and fulfilling the easy-access requirement.



4. **Demonstrate that DNS resolution is working correctly.**

Executed the `nslookup www.campus.edu` command from the PC command prompt. The output explicitly shows the request being handled by the configured DNS Server (`192.168.10.11`) and successfully returning the Web Server's

translated IP address (192.168.10.12), definitively verifying the DNS infrastructure is fully operational.

